Smart Factory Cloud and Data Infrastructure: Final Report (ME 217 Assignment 4)

ME 217 Industrial Data Analytics · 28 Oct 2026 · Group members: Jashwanth, Aksah, Ayush, Tulasi, Kushal, Benedict, Hoysala

1. Summary and recommendation

Recommendation: Microsoft Azure, with a weighted score of 4.15 out of 5, against 3.95 for AWS and 3.35 for Google Cloud. The 0.20 margin over AWS is real but narrow, so the report states what it rests on and where it would flip (Section 4).

The scenario is 100 CNC machines in three plants, producing about 15.2 GB a day of raw data if one 100 Hz vibration sensor per machine is added to the ten 1 Hz streams (Section 2). Azure fits because of three documented facts:

- **OT security.** Microsoft Defender for IoT has an actively released OT network sensor (version 26.1.1, June 2026) [R7]. The AWS page opened does not describe OT-network monitoring, and Google has no first-party IoT device service since IoT Core was retired on 16 August 2023 [R19].
- **A single path from machine to dashboard.** Azure IoT Operations reads OPC UA and runs an MQTT broker at the plant [R1], its data flows can write to Event Hubs, Azure Data Lake Storage and Azure Data Explorer [R3], and Power BI reads Data Explorer in DirectQuery mode [R5].
- **Edge reduction leaves a small cloud stream.** Feature extraction at the plant cuts 15.2 GB a day to 0.76 GB a day (95%), so cloud choice is driven by fixed costs and operations, not volume.

Three costs of this choice are stated openly. Each plant must run an Arc-enabled Kubernetes cluster [R1, R2]. Azure IoT Operations is billed per node-hour plus Device Registry per asset [R4]. MTConnect is not named among the connectors on Microsoft's overview page [R1], so a CNC that speaks only MTConnect needs an adapter. If the company is not Microsoft-centred, or cannot staff plant-level Kubernetes, AWS is a close second.

2. Part A: Industrial data volume

One byte assumption is used everywhere: 16 bytes per measurement (8-byte value + 8-byte timestamp), decimal units, 24 h a day, 30-day month, 365-day year. This is raw payload only, so real database storage is larger.

Table A1. Fleet of 100 machines, 10 sensors each at 1 Hz (the brief's baseline)

Period	Measurements	Raw size
Second	1,000	16 KB
Hour	3,600,000	57.6 MB
Day	86,400,000	1.382 GB
Month	2,592,000,000	41.47 GB
Year	31,536,000,000	504.6 GB

Table A2. One vibration sensor at 100 Hz (the brief's second calculation)

Period	Measurements	Raw size
Second	100	1.6 KB
Hour	360,000	5.76 MB
Day	8,640,000	138.2 MB
Month	259,200,000	4.147 GB
Year	3,153,600,000	50.46 GB

Table A3. Extension: one 100 Hz vibration sensor on every machine (100 sensors)

Period	Measurements	Raw size
Second	10,000	160 KB (1.28 Mbit/s)
Day	864,000,000	13.82 GB
Year	315,360,000,000	5.046 TB

A 100 Hz sensor produces as much data as 100 slow sensors, so ten such sensors equal the whole 1,000-sensor baseline. With both streams the fleet produces 950.4 million values (15.2 GB) a day and 346.9 billion (5.55 TB) a year.

Table A4. What the edge sends instead

Stream	Policy	Values a second	GB a day
Vibration	5 features (RMS, peak, crest factor, kurtosis, dominant frequency) per 1 s window per machine	500	0.691
Slow sensors	mean, min, max per minute	50	0.069

Stream	Policy	Values a second	GB a day
Total sent		**550**	**0.760**
Raw alternative		11,000	15.21

The reduction is 95.0%. At about 0.25 GB a day per plant, this is a small stream for any WAN link.

Why high-frequency vibration is hard to store and send. Volume scales with sampling rate, runs continuously, and 100 Hz is already low for diagnostics: by the Nyquist-Shannon theorem the rate must exceed twice the highest frequency of interest [R22], so bearing and gear faults normally need kilohertz sampling. At 10 kHz the same sensors would produce 100 times the volume of Table A3.

Why raw data is not always sent to the cloud. Bandwidth and per-message ingestion cost grow with volume; most healthy-machine samples carry little information; protective actions cannot wait for a WAN round trip; and the plant must keep running during an outage. Raw data is still kept in a short local buffer and a raw snippet is uploaded when an alarm fires.

Role of edge processing. The edge gateway reads machine tags, filters noise, extracts features, raises local alarms, buffers during outages and sends only compact data outward (Section 8).

Compression, aggregation and feature extraction are different tools. Compression re-encodes the same samples in fewer bytes; on noisy vibration it helps little beyond storing fewer bits per sample, and cost still scales with sampling rate. Aggregation replaces samples by summary statistics, which suits slow variables; min and max are kept so short spikes are not averaged away. Feature extraction computes diagnostic indicators and gives the largest saving, 95% here, rising with sampling rate.

3. Part B: Cloud platform comparison

All statements below were checked against vendor pages in October 2026. Services change quickly, so two recent retirements are called out.

Table B1. Twelve-criterion comparison

Criterion	AWS	Azure	Google Cloud
1. Plant-floor connectivity (OPC UA, MQTT, MTConnect)	SiteWise Edge reads OPC UA, Modbus TCP and EtherNet/IP natively; other protocols through partner software [R10]	IoT Operations has an OPC UA connector and an MQTT broker; MTConnect is not named on the overview page [R1]	Manufacturing Connect, sold and supported by Litmus, lists 270+ protocols [R16, R17]
2. Edge runtime	Greengrass v2 on a gateway [R10, R15]	Arc-enabled Kubernetes cluster at each plant; up to 72 h offline operation [R1, R2]	Litmus edge software; Google's own IoT Core was retired in August 2023 [R19]
3. Device and message ingestion	IoT Core, $1 per million messages in 5 KB steps [R14]	IoT Operations MQTT broker, data flows to Event Hubs [R3]	Pub/Sub, with no first-party device registry
4. Time-series storage	Timestream for InfluxDB; LiveAnalytics is closed to new customers [R11]	Azure Data Explorer (KQL) [R6]	BigQuery or Bigtable (general-purpose)
5. Stream processing	Kinesis, managed Flink	Stream Analytics [R8], Fabric	Dataflow
6. Machine learning for maintenance	SiteWise anomaly detection (CNC supported) [R12]; Lookout for Equipment discontinued 7 Oct 2026 [R13]	Azure ML [R9]; AI Anomaly Detector retired 1 Oct 2026 [R18]	Vertex AI
7. OT network security	Device Defender and IoT security services; no OT-network sensor found	Defender for IoT OT sensor 26.1.1 [R7]	Cloud-side security only; third-party OT tools
8. Dashboards	Grafana (managed) and QuickSight	Power BI over Data Explorer (DirectQuery) [R5]	Looker
9. Enterprise integration	Broad, neutral to ERP vendor	Entra ID, Microsoft 365 and Dynamics are native	Google Workspace, BigQuery-centred
10. Scalability and reliability	Mature, multi-region	Mature, multi-region	Mature, multi-region
11. Cost model	Per message, per asset and per GB; no required plant cluster	Per node-hour plus Device Registry per asset [R4]; plant cluster required	Per usage, plus Litmus licence
12. Operational simplicity	Gateway software plus managed services	Kubernetes at each plant to maintain	Depends on Litmus and several services

What the table means in practice.

Connectivity. All three can read an OPC UA server. The difference is who owns the connector. On AWS and Azure the connector is a cloud-vendor product; on Google it is a partner product, so support and updates come from Litmus. A CNC

that speaks only MTConnect needs an adapter on any platform, because MTConnect is not named in the native lists of AWS or Azure.

Storage and analytics. Timestream for LiveAnalytics is closed to new customers [R11], so the AWS design uses Timestream for InfluxDB.

Machine learning. Both AWS Lookout for Equipment and Azure AI Anomaly Detector are gone as of October 2026 [R13, R18]. Neither vendor therefore offers a ready-made service for this job except AWS's built-in SiteWise anomaly detection, which the vendor says suits stationary signals with limited variability [R12]. In both clouds the team would build the model itself (Azure ML or SageMaker), so the two score the same in Part C.

Security. This is the clearest difference. Azure has a maintained OT sensor that watches industrial protocols passively [R7]; no equivalent was found on the AWS or Google pages read. A plant with an IT/OT boundary to defend gains most from this.

Cost. No total was calculated with vendor price calculators, because the fleet's configuration is hypothetical. Only pricing models are compared. Azure's per-node and per-asset charges [R4] and its mandatory plant cluster make it the costlier model on fixed costs; AWS's per-message price [R14] is cheap when the edge sends 550 values a second, as in Table A4.

4. Part C: Weighted decision matrix

The twelve comparison points of Part B are merged into ten scoring criteria (items 3, 4 and 5 become one ingestion and time-series criterion; the other items stay separate). Weights reflect this factory's priorities: getting data off CNC controllers, processing it at the plant, and storing and analysing it come first because everything else depends on them.

Table C1. Criteria, weights and reasons

#	Criterion	Weight	Why this weight
1	Industrial connectivity	15%	No data, no project
2	Edge computing	15%	Three plants, safety and outage tolerance (Section 8)
3	Ingestion and time-series analytics	15%	Core of the cloud workload
4	OT security	10%	Connecting machines to a cloud widens the attack surface
5	Machine learning for maintenance	10%	The business goal, but it can be built on any platform
6	Dashboards	10%	Operators and managers see results here
7	Cost model	10%	Fixed costs matter at 100 machines
8	Enterprise integration	5%	Matters only if ERP or identity is Microsoft
9	Scalability and reliability	5%	All three are mature
10	Operational simplicity	5%	Small team

Rubric. 5 = native, documented and actively maintained, with no extra product needed; 4 = fully capable, with a minor gap or a partner step; 3 = workable with significant custom work or a third party; 2 = major gap; 1 = not available. A score is awarded only for what a vendor page states.

Table C2. Scores with evidence (S = score)

#	AWS	Azure	GCP
1	4: native OPC UA, Modbus, EtherNet/IP; MTConnect via partner [R10]	4: OPC UA connector and MQTT broker; MTConnect not named [R1]	3: protocols supplied by Litmus, a partner [R16]
2	4: Greengrass gateway [R15]	4: Arc cluster, 72 h offline [R1]; Kubernetes is a burden (see 10)	3: partner edge software; Google IoT Core retired [R19]
3	4: IoT Core plus Timestream for InfluxDB [R11, R14]	5: one path from broker to Data Explorer and KQL [R3, R6]	4: Pub/Sub and BigQuery, but no device layer
4	4: AWS security services, no OT sensor found	5: Defender for IoT OT sensor 26.1.1 [R7]	3: cloud-side only
5	4: SiteWise anomaly detection plus SageMaker; Lookout gone [R12, R13]	4: Azure ML; Anomaly Detector gone [R9, R18]	4: Vertex AI
6	4: Grafana, QuickSight	4: Power BI DirectQuery [R5]	3: Looker needs more set-up
7	4: per-message pricing, no plant cluster [R14]	3: node-hour and per-asset fees [R4]	3: usage plus Litmus licence
8	3: neutral	4: Entra ID, Microsoft 365	3: Workspace-centred
9	5	5	5
10	3: gateway plus several services	3: cluster per plant to run	3: several vendors

Table C3. Weighted totals

Platform	Total (out of 5)
Azure	**4.15**
AWS	3.95

Platform	Total (out of 5)
Google Cloud	3.35

Example (Azure): 0.15×4 + 0.15×4 + 0.15×5 + 0.10×5 + 0.10×4 + 0.10×4 + 0.10×3 + 0.05×4 + 0.05×5 + 0.05×3 = 4.15.

Azure leads AWS in only three criteria (ingestion and time series, OT security, enterprise integration) and trails it in one (cost); the other six are ties. That is why the margin is small.

Table C4. Does the ranking hold?

Change	AWS	Azure	GCP
Base case	3.95	4.15	3.35
Equal weights (10% each)	3.90	4.10	3.40
Azure ingestion score 5 to 4	3.95	4.00	3.35
Azure security score 5 to 4	3.95	4.05	3.35
Azure cost score 3 to 2	3.95	4.05	3.35
Azure edge 4 to 3 and simplicity 3 to 2 (heavy Kubernetes burden)	3.95	3.95	3.35
Four Azure drops together (ingestion, security, enterprise, cost)	3.95	3.75	3.35

Reading the table honestly. Azure's lead survives equal weights and any single score change tested, and ties AWS when the Kubernetes burden is scored harshly. It does not survive several simultaneous drops (last row). A simulation that moved every score at random by up to ±1 point gave Azure the highest total in about 60% of trials, AWS in about 39% and Google in about 1%. AWS draws level with Azure if the cost weight rises from 10% to 25% (other weights scaled down in proportion) and overtakes above that. Google Cloud is clearly third in every test.

Scenario-based recommendation. Choose Azure if OT security monitoring matters and a Microsoft-centred IT environment (or at least staff able to run plant clusters) exists. Choose AWS if the team wants the lowest fixed cost, has no Kubernetes skills, or already runs on AWS. Choose Google only if it is already the company's data platform and Litmus is acceptable as the plant layer. The scores are the authors' judgement against the rubric; they are not measurements, and a different group could reasonably move individual scores by one point.

5. Part D: SQL analysis of machine data

Data. A seeded simulator generated 100,800 rows: 10 machines (M01–M10), one row per machine per minute, 7 days from 22 Oct 2026. The data is synthetic, so results show that the queries work, not how a real plant behaves. Table `manufacturing_data` has Record_ID, Machine_ID, Timestamp, Temperature, Vibration, RPM, Current, Production_Count, Rejected_Count, Machine_Status (Running, Idle, Stopped, Fault) and Alarm (0/1), with CHECK constraints on status, alarm and counts, and indexes on (Machine_ID, Timestamp) and Timestamp.

Two meaning decisions:

- Production_Count and Rejected_Count are **per-minute increments**, not running totals, so totals are SUMs. Summing a running counter would overcount.
- Production_Count **includes** rejected parts, so rejection rate = SUM(Rejected_Count) / SUM(Production_Count), and the schema enforces Rejected_Count ≤ Production_Count.

For "highest" questions the queries use `RANK()`, so a tie returns every tied machine instead of silently dropping one. Each Running row is one minute, so Running_Hours = row count / 60.

Table D1. Queries and results on the test data

Query	Question	Method	Result
Q1	Average temperature per machine	AVG, GROUP BY	56.44 °C (M05) to 62.35 °C (M08); all ten are listed in the query output
Q2	Machine with most parts	SUM, RANK	M10, 7,248 parts
Q3	Highest average vibration	AVG, RANK	M02, 2.07
Q4	Alarms per machine	SUM(Alarm)	M05 713, M02 554, M07 469, M01 454, M04 415, M08 365, M03 322, M10 320, M09 295, M06 223
Q5	Parts per machine on 26 Oct 2026, then plant total	Date range, SUM	Plant total 10,092; highest M09 with 1,188
Q6	Running time per machine	COUNT of Running rows	M10 119.02 h (7,141 min) highest; M07 100.83 h lowest
Q7	Average RPM while running	AVG with filter	1,448.7 (M06) to 1,449.6 (M03, M04); overall 1,449.3
Q8	Highest rejection rate	SUM ratio, RANK	M05, 441 of 6,360 = 6.93%

Q5 uses a half-open range (`>= 2026-10-26 00:00:00 AND < 2026-10-27 00:00:00`) so every minute of the day is counted exactly once and an index on Timestamp can be used.

Testing statement. The queries were run in SQLite 3 on the generated database and are written for MySQL 8 (the CTEs and RANK need version 8.0+). They have not been run on a MySQL server, and the schema's AUTO_INCREMENT is converted to AUTOINCREMENT by the loader for SQLite. The complete SQL is in the repository (Section 6).

6. Part E: GitHub repository

Repository link: https://github.com/venkatjashwanth/industrial-data-platform

Structure:

```
industrial-data-platform/
  data/         factory.db (generated SQLite test database, 100,800 rows)
  python/       generate_data.py (seeded simulator, seed 217)
  sql/          schema.sql, queries.sql (Q1-Q8)
  dashboard/    rejection-rate chart built from query Q8 (PNG)
  diagrams/     F1_azure_architecture.png, G1_hybrid_edge_cloud.png
  README.md     scenario, dataset, how to run, SQL instructions, findings
  requirements.txt
  report.pdf    this report
```

The README states what the data is (synthetic, 10 machines, 7 days), how to rebuild it and run the queries, that counters are per-minute increments, and that testing was done in SQLite. Commit messages say what changed and why (for example "Fix Q5 date range to half-open"). This report, exported as PDF, is report.pdf in the repository.

7. Part F: Industrial cloud architecture on Azure

Design basis. 100 CNC machines in three plants (about 33 each) produce 15.2 GB a day of raw data. The edge sends 0.76 GB a day, about 0.25 GB a day per plant (Table A4). Four principles follow from that:

- The plant network (OT) keeps its machines; one gateway per plant is the only outbound path.
- Data is reduced at the plant, before it crosses the WAN.
- Each tier of storage has one job: raw history in a cheap lake, recent readings in a time-series store, curated results in SQL.
- People approve any command sent back to a machine.

Figure F1. Azure architecture for the smart factory, from machines (1) to manufacturing decision (9).

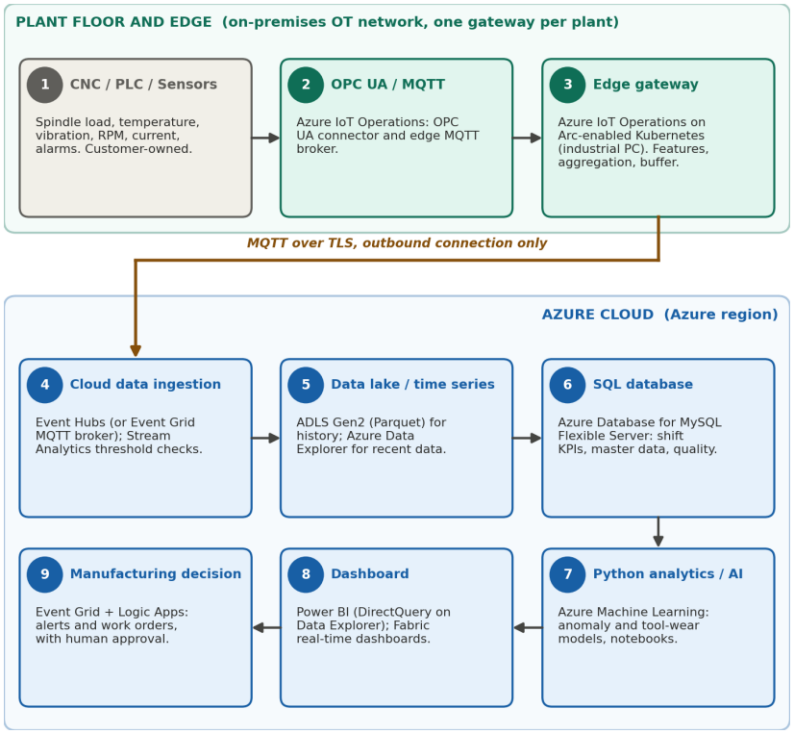


Table F1. The nine architecture blocks mapped to Azure.

#	Block	Azure component	What it does in this factory
1	CNC / PLC / sensors	None (customer-owned)	Spindle load, temperature, vibration, RPM, current, part counts and alarms. Identical on every cloud, so it does not separate the platforms.
2	OPC UA / MQTT	Azure IoT Operations: connector for OPC UA, edge MQTT broker [R1]	The connector reads controller tags and publishes them to MQTT topics at the plant. A CNC that speaks only MTConnect needs an adapter, because MTConnect is not named in the connector list [R1].
3	Edge gateway	Azure IoT Operations on an Arc-enabled Kubernetes cluster (K3s on Ubuntu 24.04, or AKS Edge Essentials on Windows) on an industrial PC in each plant [R1, R2]	Data flows normalise and aggregate tags. A custom container computes the vibration features (customer workloads can run on the platform [R1]). Operation continues offline for up to 72 hours, with possible degradation [R1].
4	Cloud ingestion	Data flow to Event Hubs (Kafka) or the Event Grid MQTT broker [R1, R3]; Azure Stream Analytics for threshold checks [R8]	Receives only features, aggregates, alarms and event snapshots. IoT Hub is not required for this path.
5	Data lake / time-series storage	Azure Data Lake Storage Gen2 (Parquet) and Azure Data Explorer or Fabric Eventhouse [R3, R6]	The lake keeps full history for audit and model training. Data Explorer serves fast time-range queries on recent data.
6	SQL database	Azure Database for MySQL Flexible Server, loaded by a scheduled job (for example Azure Data Factory)	Machine master data, shift-level OEE, downtime reasons, quality results and the Part D tables. MySQL keeps the Part D queries unchanged; Azure SQL Database would need a T-SQL rewrite.
7	Python analytics / AI	Azure Machine Learning notebooks and endpoints [R9]	Anomaly and tool-wear models built with pandas and scikit-learn. The Azure AI Anomaly Detector service was retired on 1 October 2026 [R18], so the team builds the model itself.
8	Dashboard	Power BI in DirectQuery mode over Data Explorer [R5]; Fabric real-time dashboards for the shop floor [R1]	Operators see live status and alerts; managers see OEE, downtime and rejection rate.
9	Manufacturing decision	Event Grid and Logic Apps	Threshold breaches and model outputs notify supervisors and create maintenance work orders in the plant's maintenance or MES system. A person approves any setpoint change.

How the data flows, with numbers.

1. **Capture.** Each plant has about 330 slow streams at 1 Hz and about 33 vibration sensors at 100 Hz, roughly 3,700 raw values a second.
2. **Reduce.** The gateway keeps raw vibration in a short local buffer and sends 5 features per 1-second window plus 1-minute mean, min and max for slow streams: about 183 values a second per plant (0.25 GB a day).
3. **Ingest.** Outbound TLS carries those values to Event Hubs. Stream Analytics checks thresholds as data arrives.
4. **Store.** The same stream lands in the lake and in Data Explorer. A raw snippet is uploaded only when an alarm fires.
5. **Curate.** A scheduled job writes shift-level results into MySQL, ready to join with work orders.
6. **Learn and act.** Azure Machine Learning trains on lake and SQL data. Alerts and work orders leave through Event Grid and Logic Apps.

Storage tiers (retention periods are assumptions, to be set with the plant). Raw history and features go to ADLS Gen2 (Parquet, partitioned by plant, machine and date) for long-term, low-cost retention. About 90 days of features and aggregates stay in Azure Data Explorer for fast queries. Curated shift summaries, master data, quality results and work orders live in MySQL for the life of the records.

Security design.

- The gateway starts every connection outward over TLS, so no inbound path from the internet reaches the plant network.
- Plant network and enterprise network are separated into zones, with the gateway as the only conduit, following IEC 62443 zone-and-conduit practice [R20] and NIST SP 800-82 guidance on OT security [R21].
- The Defender for IoT OT network sensor watches plant traffic passively; version 26.1.1 was released in June 2026 [R7].
- Access is role-based through Entra ID, and device and service credentials are held in a secrets store.

Known gaps. Each plant must run and patch an Arc-enabled Kubernetes cluster, and only the latest three minor versions of Azure IoT Operations receive security patches [R1, R2]; one standard gateway image and staged upgrades (one plant first) limit the effort. Controllers that speak only MTConnect need an adapter, so protocols should be inventoried before purchase [R1]. Offline operation is limited to 72 hours, so local disk must be sized for the buffer [R1]. There is no ready-made anomaly service [R18], so models are built in Azure Machine Learning, starting with simple RMS and kurtosis thresholds. Node-hour and per-asset charges [R4] should be priced in the Azure calculator before approval. The first deployment is limited to alerts and recommendations; write-back to machines comes only after the models are validated on real data.

8. Part G: Edge versus cloud decision

Table G1. Where each operation runs.

Operation	Layer	Azure component	Why
High-frequency vibration processing	Edge	Feature container on the plant cluster	100 Hz raw data stays at the plant; only features travel (Table G2).
Immediate alarms	Edge	Threshold rules in the gateway; HMI or andon light	Must fire within the plant, even when the WAN is down.
Local machine protection	Edge, final logic in the PLC	CNC controller or safety controller	Deterministic response without a network. The gateway supports the PLC; it does not replace it.
Noise filtering	Edge	Gateway data flow or container	Cleans data once at the source and cuts false alarms.
Data aggregation	Edge	Gateway data flow	1-second values become 1-minute mean, min and max.
Store-and-forward buffering	Edge	Gateway local storage	Keeps data during an outage and sends it when the link returns.
Historical analysis	Cloud	Data Explorer and ADLS Gen2	Months of history show wear and drift.
Fleet-level analysis	Cloud	Data Explorer, Power BI	One place to compare 100 machines and three plants.
Long-term storage	Cloud	ADLS Gen2	Cheap, durable, replayable for audits and retraining.
AI/ML model development	Cloud	Azure Machine Learning	Large datasets and compute for training. Trained models run at the edge for inference.
Management dashboards	Cloud	Power BI	One consistent view from anywhere, with Entra ID access.

Safety boundary. Hard real-time protection stays in the CNC controller or safety PLC. A Kubernetes gateway on an industrial PC is not deterministic, so the gateway raises alerts and requests a feed hold through the PLC; it is not the last line of defence.

Figure G1. Hybrid edge-cloud data flow: what runs at the plant, what runs in Azure, and the model update loop.

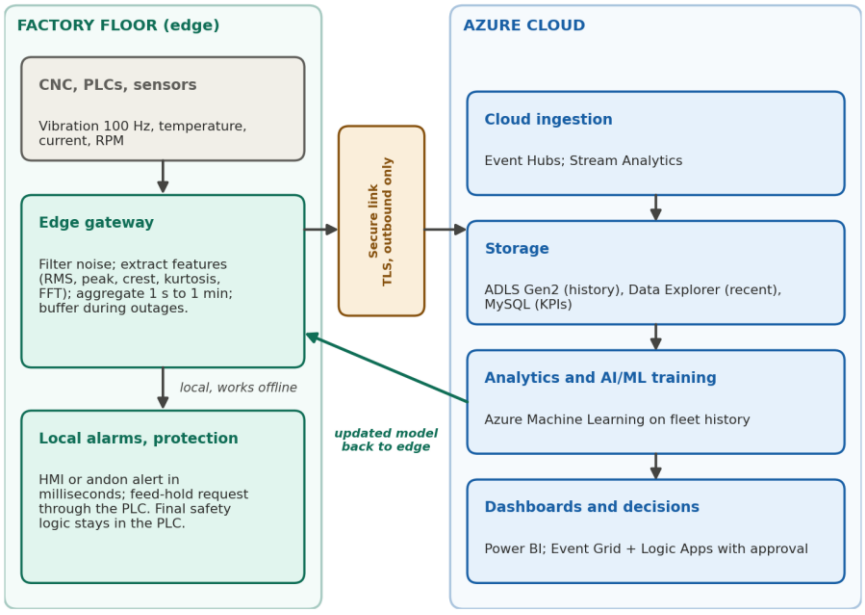


Table G2. Bandwidth with and without edge processing (payload only, 16 bytes a value, from Table A4).

Case	Values a second	All plants (kbit/s)	Per plant (kbit/s)	GB a day

All raw to cloud (100 Hz vibration)	11,000	1,408	about 469	15.21
Edge-processed to cloud	550	70.4	about 23.5	0.760
All raw, vibration at 10 kHz (illustrative)	about 1,001,000	about 128,000	about 42,700	about 1,384
Edge-processed, vibration at 10 kHz	550	70.4	about 23.5	0.760

Real messages carry headers and formatting, so network load is several times the payload figures. Batching reduces but does not remove that overhead. The last two rows show the main scaling argument: raw volume grows with sampling rate, while the feature stream stays the same at 550 values a second.

Engineering justification by criterion.

- **Latency.** Local processing responds in milliseconds with predictable timing. A WAN round trip to the cloud takes tens to hundreds of milliseconds and varies (assumed typical range, not measured here). Alarms and protection therefore run at the plant; historical reports tolerate delay and run in the cloud.
- **Reliability.** The gateway keeps alarming and buffering during an outage, within Azure IoT Operations' stated 72-hour offline limit [R1]. The cloud provides durable storage and backup, so the plant keeps operating and the cloud keeps the record.
- **Bandwidth.** Raw vibration alone is about 1.4 Mbit/s for the fleet at 100 Hz; features cut it to about 70 kbit/s, a 95% reduction (Table G2). Three plants sharing WAN links with ERP and MES traffic should not carry raw streams continuously.
- **Cost.** Sending raw data raises ingestion, storage and egress with volume; processing at the edge cuts the number of values sent by a factor of 20 (11,000 to 550 a second). The offset is the fixed cost of a gateway and cluster in each plant, plus node-hour and per-asset charges [R4]. No total is calculated here because the real configuration is not known; price it in the Azure calculator.
- **Cybersecurity.** The edge keeps raw process data inside the plant and limits what leaves it, with outbound-only TLS connections and a single conduit between zones [R20, R21]. The cost is more devices to patch: previous minor versions of Azure IoT Operations receive no security updates [R2]. The cloud adds managed identity and monitoring.
- **Scalability.** A new plant gets the same gateway image and templates; cloud storage and compute grow with software settings, not hardware. Per-asset registry charges grow with machine count [R4].

Edge disadvantages to state openly. Edge hardware is limited in compute and storage, must be bought and installed at every plant, faces heat, dust and vibration, and sees only its own plant. These are the reasons the cloud keeps the fleet view and the training workload.

9. Part H: Final recommendation and conclusion

Recommendation: Microsoft Azure with a hybrid edge-cloud design. Azure scored 4.15 out of 5, against 3.95 for AWS and 3.35 for Google Cloud (Table C3). The margin is narrow, and Section 4 shows where it holds and where it fails. The recommendation rests on the scenario, not only on the score:

- The architecture has one documented path from the OPC UA connector to the dashboard (Table F1), with a maintained OT security sensor [R7] and Power BI reading Data Explorer directly [R5].
- Edge reduction makes the cloud stream small (0.76 GB a day), so the choice is decided by fixed costs and plant operations, not by data volume.

When to choose differently. Choose AWS if the team has no Kubernetes skills, wants the lowest fixed cost, or already runs on AWS. Choose Google Cloud only if it is already the company's data platform and Litmus is acceptable as the plant layer [R16, R17].

Suggested rollout. Pilot: one plant, about 10 machines, read-only, alerts to a dashboard, to check protocols (including MTConnect), data volume and alarm accuracy. Extend: all three plants with the standard gateway image, SQL load jobs and shift KPIs. Close the loop: maintenance work orders raised automatically; any machine write-back only with human approval.

Limitations. Scores are the authors' judgement against a rubric, not measurements. Data volumes use a 16-byte payload and 24-hour operation, so they are upper-bound payload figures. No price was calculated because the configuration is hypothetical. The SQL data is synthetic and the queries were tested in SQLite, not on a MySQL server. Network latency figures are assumed typical ranges, not measurements. No hardware or cloud subscription was used; the architecture is a design, not a tested deployment.

Conclusion. The company needs real-time protection at the plant and fleet-wide analysis in the cloud, and neither layer can do the other's job. Reducing data at the edge makes the cloud workload small and cheap to run, while the cloud supplies storage, model training and a single view of three plants. Azure fits this scenario best because it offers the shortest documented path from OPC UA machines to dashboards and an OT security monitor, at the price of running a Kubernetes cluster in each plant.

Table H1. Where each required deliverable is in this report.

Deliverable	Location
Data-volume calculations	Section 2 (Tables A1 to A4)
Cloud comparison table	Section 3 (Table B1)
Weighted decision matrix	Section 4 (Tables C1 to C4)
SQL schema and queries	Section 5 and the repository
GitHub repository link	Section 6 (link given there)
Cloud architecture diagram	Section 7 (Figure F1, Table F1)
Edge-versus-cloud analysis	Section 8 (Figure G1, Tables G1, G2)
Final platform recommendation	Sections 1 and 9
Technical report	This document

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